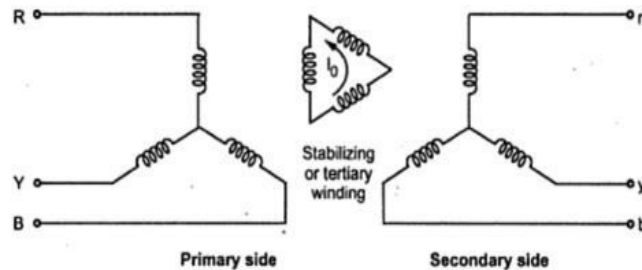


THREE WINDING TRANSFORMER

- In some high rating transformers, a third winding is constructed in addition to the primary and secondary windings. This third winding is called Tertiary Winding and the transformer is called Three Winding Transformer.
- The voltage ratings of all three windings of transformer are usually unequal. The primary winding has the highest voltage rating; the tertiary has the lowest voltage rating, and the secondary has the intermediate voltage rating.
- The most significant advantage of the third winding is that the harmonic generated by the primary and secondary winding extinguish by the third winding. The third winding is connected in delta.



Various Purposes of Tertiary Windings

- It reduces the unbalancing in the primary due to three phase unbalanced load.
- It can supply additional load providing isolation from the secondary windings.
- It can act as a voltage source to supply substation auxiliaries at a voltage different from those of primary and secondary winding.
- Three transmission lines at different voltage levels can be interconnected by using three winding transformer.
- Tertiary winding can be served for the purpose of measuring voltage of a high voltage transformer.
- Static capacitors or synchronous condensers may be connected to the tertiary winding for reactive power injection into the system.
- A delta-connected tertiary reduces the impedance offered to the zero sequence currents thereby allowing a large earth-fault current to flow through the protective device facilitating its proper operation.
- It also permits the third harmonic current to flow.